

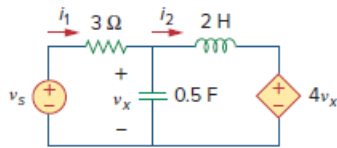
Problem

For the circuit in Fig. 16.97, find:

(a) I_1/V_s

(b) I_2/V_x

Figure 16.97:



Step-by-step solution

Step 1 of 7

Refer the Figure 16.97 in the textbook for the actual circuit.

(a)

Consider the circuit shown below

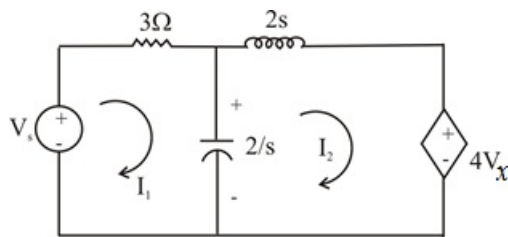


Figure 1

Step 2 of 7

Apply Kirchhoff's voltage law to loop 1.

$$V_s = \left(3 + \frac{2}{s}\right) I_1 - \frac{2}{s} I_2 \quad \dots\dots (1)$$

Apply Kirchhoff's voltage law to loop 2.

$$4V_x + \left(2s + \frac{2}{s}\right) I_2 - \frac{2}{s} I_1 = 0$$

From the circuit, the voltage

$$V_x = (I_1 - I_2) \frac{2}{s}$$

Substitute $(I_1 - I_2) \frac{2}{s}$ for V_x .

$$4 \left((I_1 - I_2) \frac{2}{s} \right) + \left(2s + \frac{2}{s} \right) I_2 - \frac{2}{s} I_1 = 0$$

$$\frac{8}{s} (I_1 - I_2) + \left(2s + \frac{2}{s} \right) I_2 - \frac{2}{s} I_1 = 0$$

$$-\frac{6}{s} I_1 + \left(\frac{6}{s} - 2s \right) I_2 = 0 \quad \dots\dots (2)$$

Step 3 of 7

Write the equations (1) and (2) in matrix form.

$$\begin{bmatrix} V_s \\ 0 \end{bmatrix} = \begin{bmatrix} 3 + \frac{2}{s} & \frac{-2}{s} \\ \frac{-6}{s} & \frac{6}{s} - 2s \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\Delta = \left(3 + \frac{2}{s}\right)\left(\frac{6}{s} - 2s\right) - \left(-\frac{6}{s}\right)\left(-\frac{2}{s}\right)$$

$$\Delta = \frac{18}{s} - 6s - 4$$

Step 4 of 7

Determine the value of Δ_1 .

$$\Delta_1 = \begin{vmatrix} V_s & \frac{-2}{s} \\ 0 & \frac{6}{s} - 2s \end{vmatrix}$$

$$= \left(\frac{6}{s} - 2s\right)V_s$$

Determine the value of Δ_2 .

$$\Delta_2 = \begin{vmatrix} 3 + \frac{2}{s} & V_s \\ \frac{-6}{s} & 0 \end{vmatrix}$$

$$= \left(\frac{6}{s}\right)V_s$$

Step 5 of 7

Calculate the current I_1 .

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{\frac{6}{s} - 2s}{\frac{18}{s} - 4 - 6s} V_s$$

$$\frac{I_1}{V_s} = \frac{\frac{3}{s} - s}{\frac{9}{s} - 2 - 3s} = \frac{s^2 - 3}{3s^2 + 2s - 9}$$

Therefore, the value of $\frac{I_1}{V_s}$ is $\boxed{\frac{s^2 - 3}{3s^2 + 2s - 9}}$.

Step 6 of 7

(b)

Determine the value of I_2 .

$$I_2 = \frac{\Delta_2}{\Delta}$$

$$V_x = \frac{2}{s}(I_1 - I_2)$$

$$= \frac{2}{s} \left(\frac{\Delta_1 - \Delta_2}{\Delta} \right)$$

$$V_x = \frac{\frac{2}{s} V_s \left(\frac{6}{s} - 2s - \frac{6}{s} \right)}{\Delta}$$

$$= \frac{-4V_s}{\Delta}$$

$$\Delta = \frac{-4V_s}{V_x}$$

Step 7 of 7

Calculate $\frac{I_2}{V_x}$.

$$I_2 = \frac{\Delta_2}{\left(\frac{-4V_s}{V_x} \right)}$$

$$\frac{I_2}{V_x} = \frac{\Delta_2}{-4V_s}$$

$$\begin{aligned} \frac{I_2}{V_x} &= \frac{\frac{6}{s} V_s}{-4V_s} \\ &= \frac{-3}{2s} \end{aligned}$$

Therefore, the value of $\frac{I_2}{V_x}$ is $\boxed{\frac{-3}{2s}}$